

Machine Learning

CMPSCI 589

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Logistic Regression

UMassAmherst
College of Information
& Computer Sciences



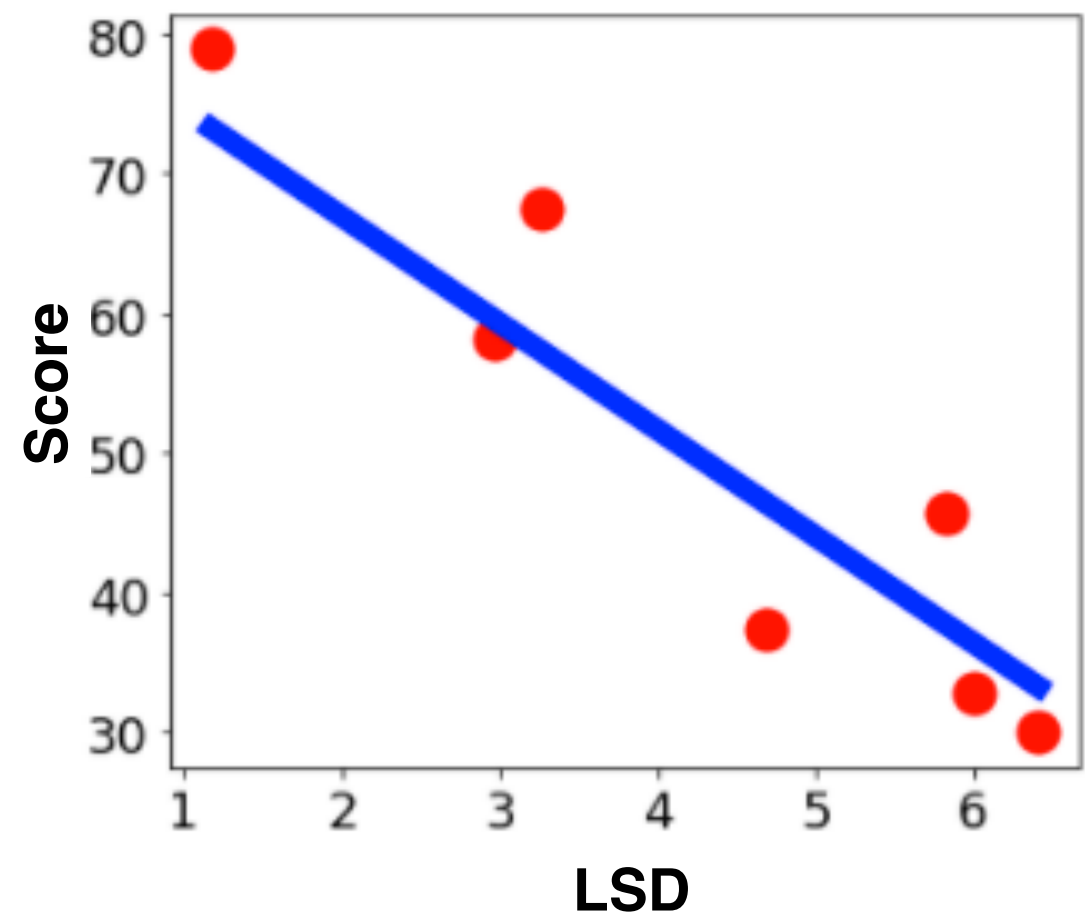
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Review: Regression Problems

- Regression algorithms: predict a real number (i.e., models a function)

	x =LSD Dosage	y = Exam Score
Student1	1.17	78.93
Student2	2.97	58.20
Student3	3.26	67.47
Student4	4.69	37.47
Student5	5.83	45.65
Student6	6.00	32.92
Student7	6.41	29.97



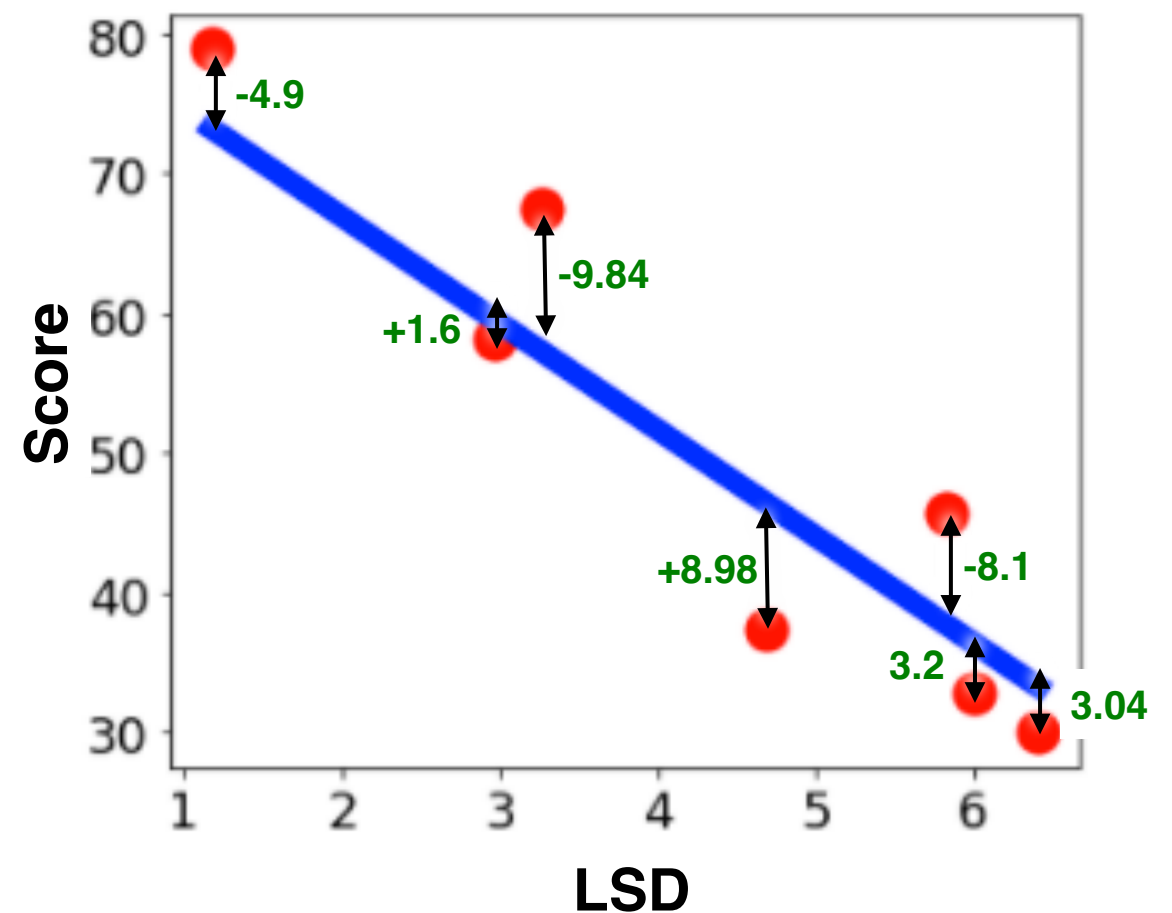
Hypothesis:
 $f(x) = \theta_0 + \theta_1 x$

Assume $\theta_0 = 83.089$
 $\theta_1 = -7.812$

Review: Regression Problems

- Regression algorithms: predict a real number (i.e., models a function)

	x =LSD Dosage	y = Exam Score	f(x) = Predicted Score
Student1	1.17	78.93	73.94
Student2	2.97	58.20	59.88
Student3	3.26	67.47	57.62
Student4	4.69	37.47	46.45
Student5	5.83	45.65	37.54
Student6	6.00	32.92	36.21
Student7	6.41	29.97	33.01



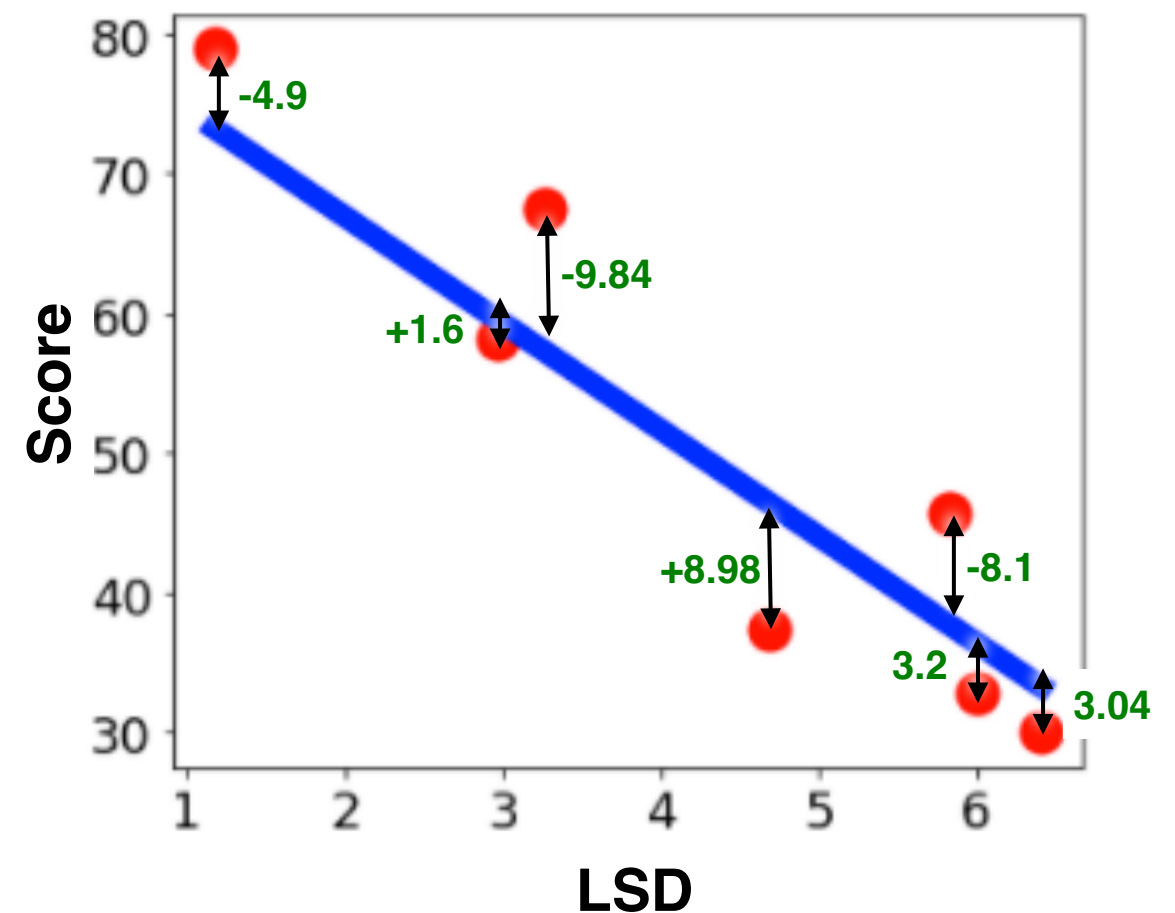
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How "wrong"
is this model?

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Assume $\theta_0 = 83.089$
 $\theta_1 = -7.812$

How “wrong”
 is this model?

$$J(\theta_0, \theta_1) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2$$

where $f_{\theta}(x) = \theta_0 + \theta_1 x$ is our hypothesis

We wish to find values for θ_0 and θ_1 that minimize the Mean Squared Error, $J(\theta_0, \theta_1)$

Review: Gradient Descent

We wish to find values for θ_0 and θ_1 that minimize the **Mean Squared Error**

$$J(\theta_0, \theta_1) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2 \quad \text{where } f_{\theta}(x) = \theta_0 + \theta_1 x \text{ is our hypothesis}$$

To decrease the error **J**, we should move each θ in the opposite direction of the derivative

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\theta_0 := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

$$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

$$\theta_1 := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

Review: Gradient Descent

We wish to find values for θ_0 and θ_1 that minimize the Mean Squared Error

$$J(\theta_0, \theta_1) = \frac{1}{2n} \sum_{i=1}^n (f_{\theta}(x^{(i)}) - y^{(i)})^2 \quad \text{where } f_{\theta}(x) = \theta_0 + \theta_1 x \text{ is our hypothesis}$$

To decrease the error J , we should move each θ in the opposite direction of the derivative

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = ?$$

$$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = ?$$

Review: Gradient Descent

Hypothesis: $f_{\theta}(x) = \theta_0 + \theta_1 x$

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = ?$$

Review: Gradient Descent

Hypothesis: $f_{\theta}(x) = \theta_0 + \theta_1 x$

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1$$

Review: Gradient Descent

Hypothesis: $f_{\theta}(x) = \theta_0 + \theta_1 x$

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1$$

$$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) x \text{ ?}$$

Review: Gradient Descent

Hypothesis: $f_{\theta}(x) = \theta_0 + \theta_1 x$

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1$$
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Review: Gradient Descent

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x^{(i)}) - y^{(i)} \right) 1$$
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$$\theta_0 := \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$$

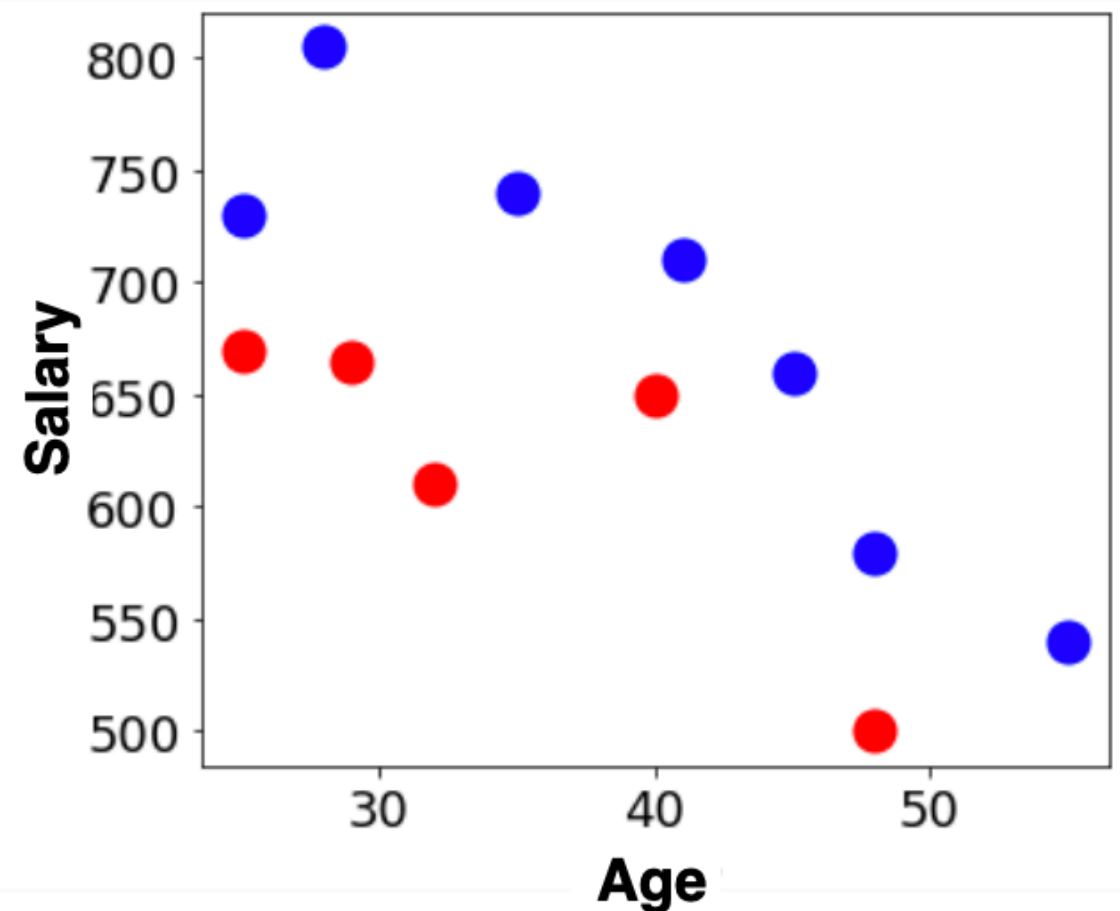
$$\theta_1 := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1)$$

Logistic Regression

Classification Problems

- Classification algorithms: predict a label (class)

	$x_1 =$ Age	$x_2 =$ Salary	$y =$ Repay Loan
Client1	25	670	No
Client2	29	665	No
Client3	32	610	No
Client4	40	650	No
Client5	48	500	No
Client6	25	730	Yes
Client7	28	805	Yes
Client8	35	740	Yes
Client9	41	710	Yes
Client10	45	660	Yes
Client11	48	580	Yes
Client12	55	540	Yes



$f(\text{Age}, \text{Salary}) \rightarrow \text{Probability of Repay Loan} = \text{Yes}$

$f(x_1, x_2) \rightarrow [0, 1]$

Hypothesis:

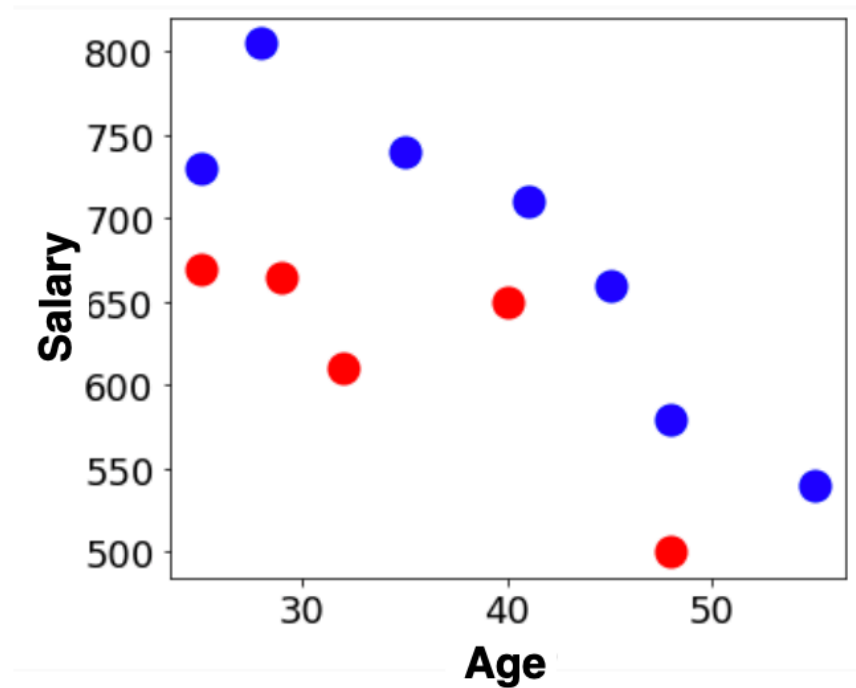
$$f(x_1, x_2) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

Problem:

- $f(x_1, x_2)$ can produce/return any values!
- Not just values between 0-1 (i.e., probabilities)

Logistic Regression

	$x_1 =$ Age	$x_2 =$ Salary	$y =$ Repay Loan
Client1	25	670	No
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$f(\text{Age, Salary}) \rightarrow \text{Probability of Repay Loan} = \text{Yes}$
 $f(x_1, x_2) \rightarrow [0, 1]$

Hypothesis:

$$f(x_1, x_2) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

Problem:

- $f(x_1, x_2)$ can produce/return any values!
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We must ensure $0 < f(x_1, x_2) < 1$

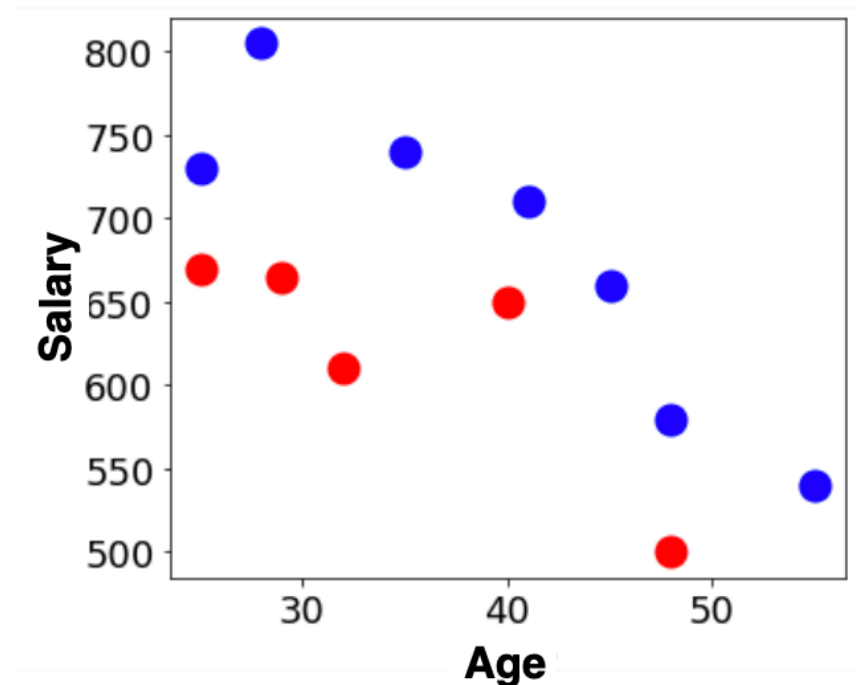
New hypothesis:

$$f(x_1, x_2) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

where $g(\cdot)$ is a function that maps arbitrary input values to the $[0-1]$ interval

Logistic Regression

	$x_1 =$ Age	$x_2 =$ Salary	$y =$ Repay Loan
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Client9	41	710	Yes
Client10	45	660	Yes
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Client12	55	540	Yes



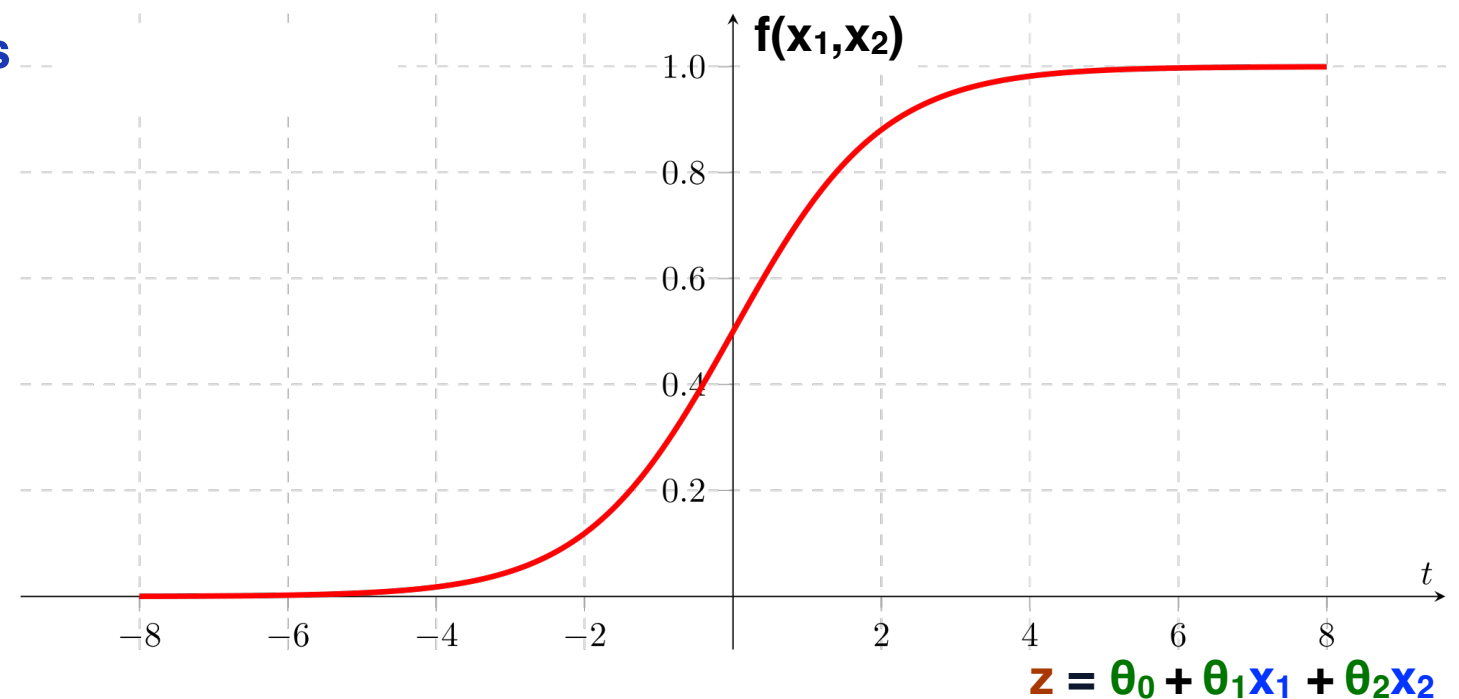
$f(\text{Age, Salary}) \rightarrow \text{Probability of Repay Loan} = \text{Yes}$
 $f(x_1, x_2) \rightarrow [0, 1]$

Hypothesis:

$$f(x_1, x_2) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

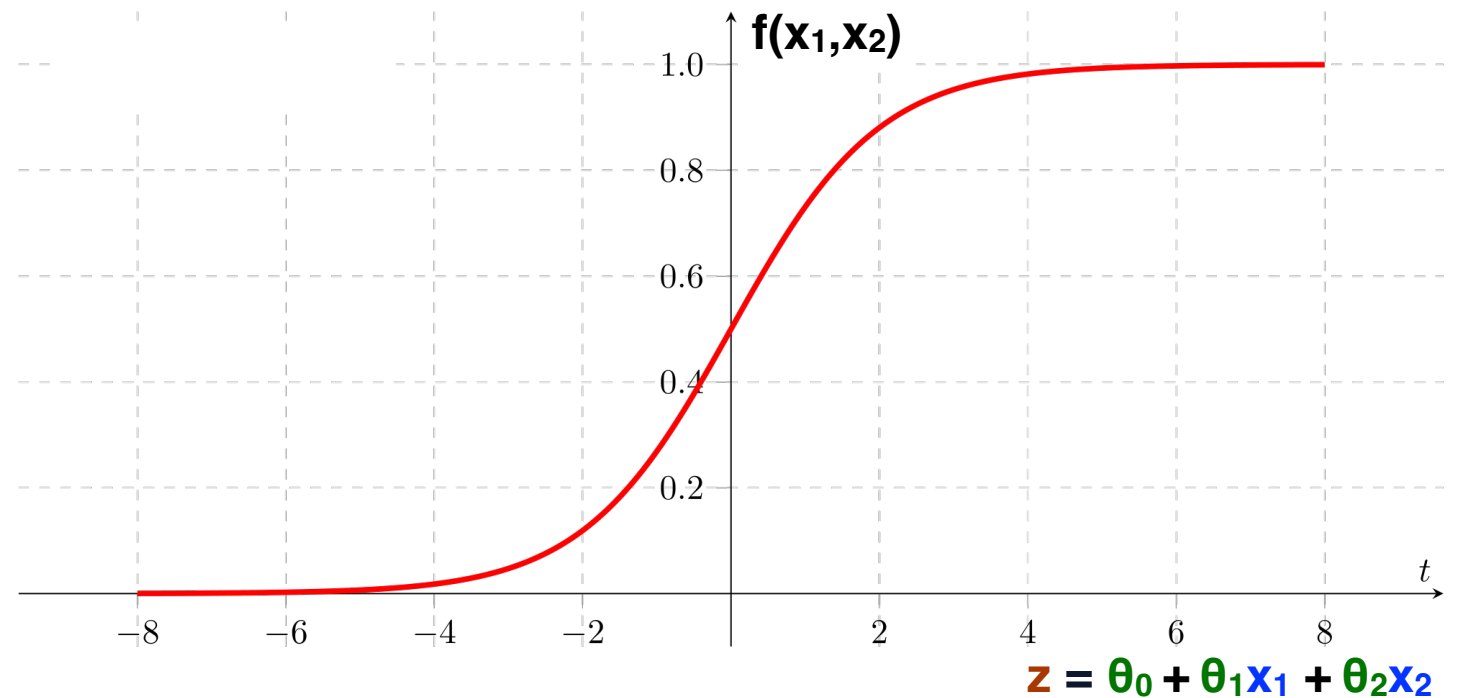
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New hypothesis:

$$f(x_1, x_2) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

where $g(.)$ is a function that maps arbitrary input values to the $[0-1]$ interval

$g(z)$ is a *sigmoid* function

$$g(z) = \frac{1}{1 + \exp(-z)}$$

$$= \frac{1}{1 + \exp\left(-(\theta_0 + \theta_1 x_1 + \theta_2 x_2)\right)}$$

Logistic Regression

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Idea:

- $f(x_1, x_2)$ is a number between 0-1
- Can be interpreted as a probability
 - probability that client will Repay Loan
- if $f(x_1, x_2) > 0.5$
 - predict $y = \text{Yes}$ (Will Repay Loan)
- if $f(x_1, x_2) \leq 0.5$
 - predict $y = \text{No}$ (Will not Repay Loan)

We must ensure $0 < f(x_1, x_2) < 1$

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When is $f(x_1, x_2) > 0.5$?

Logistic Regression

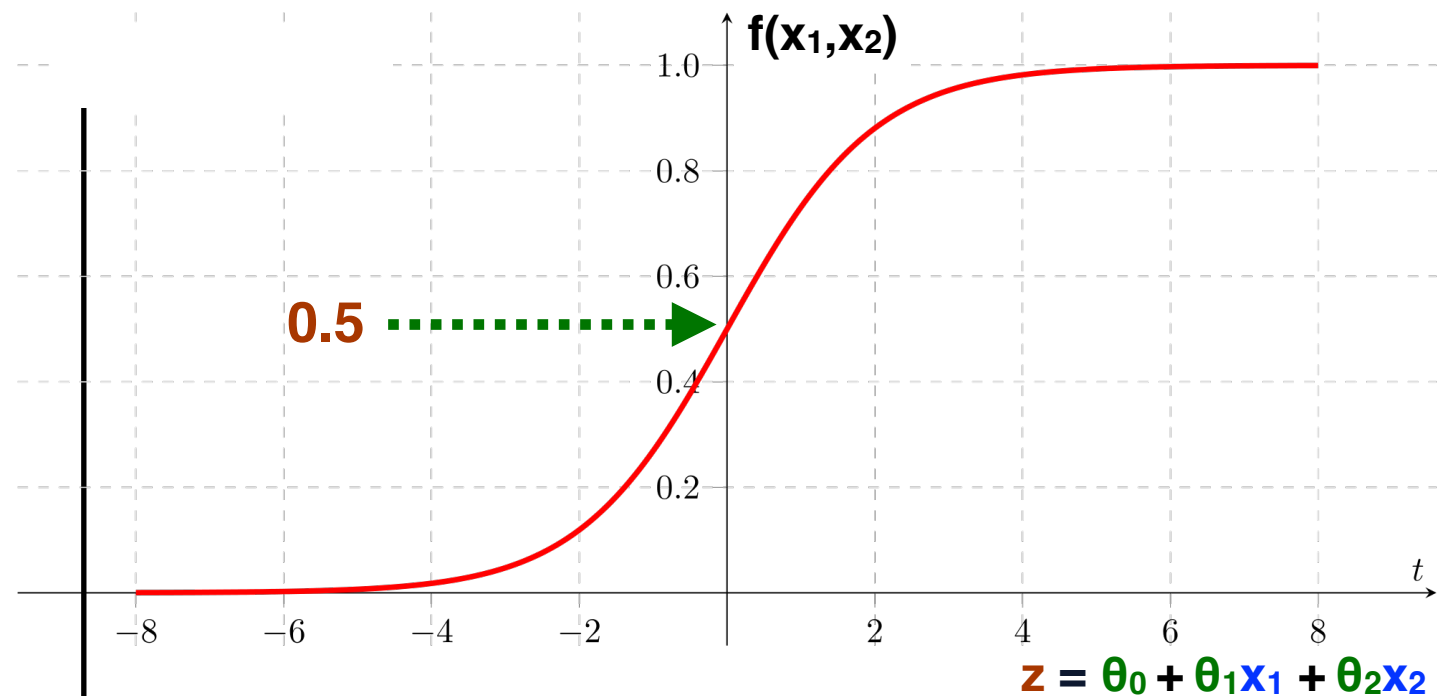
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When is $f(x_1, x_2) > 0.5$?

Whenever $\theta_0 + \theta_1 x_1 + \theta_2 x_2 > 0$

Logistic Regression

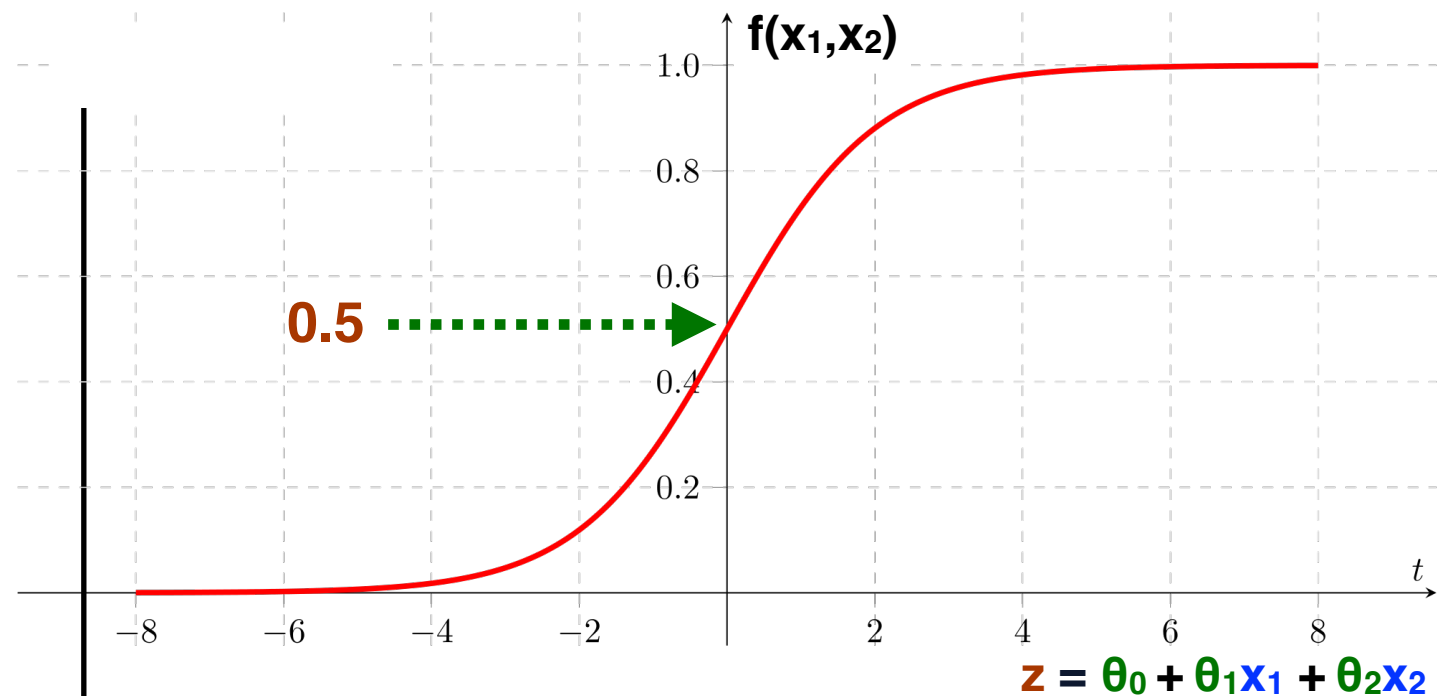
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When is $f(x_1, x_2) > 0.5$?

Whenever $\theta_0 + \theta_1 x_1 + \theta_2 x_2 > 0$

Logistic Regression - Decision Boundary

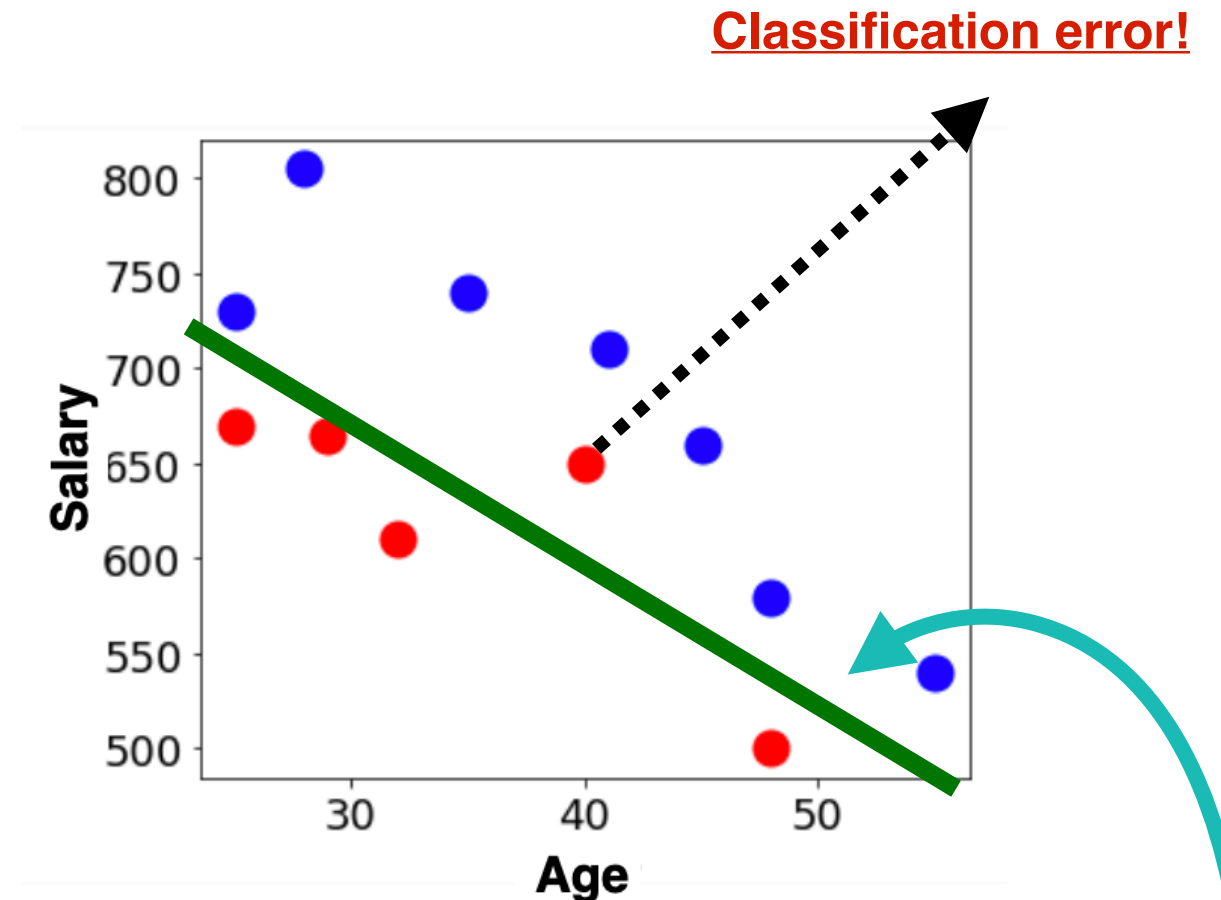
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 - predict $y=\text{No}$ (Will not Repay Loan)



Decision boundary

- separates instances with predictions $y=\text{Yes}$ and $y=\text{No}$
- the surface containing all points (x_1, x_2)
 - for which $\theta_0 + \theta_1 x_1 + \theta_2 x_2 = 0$

If $\theta_0 = -3.55$, $\theta_1 = 3.75$, $\theta_2 = 4.82$

Logistic Regression - Decision Boundary

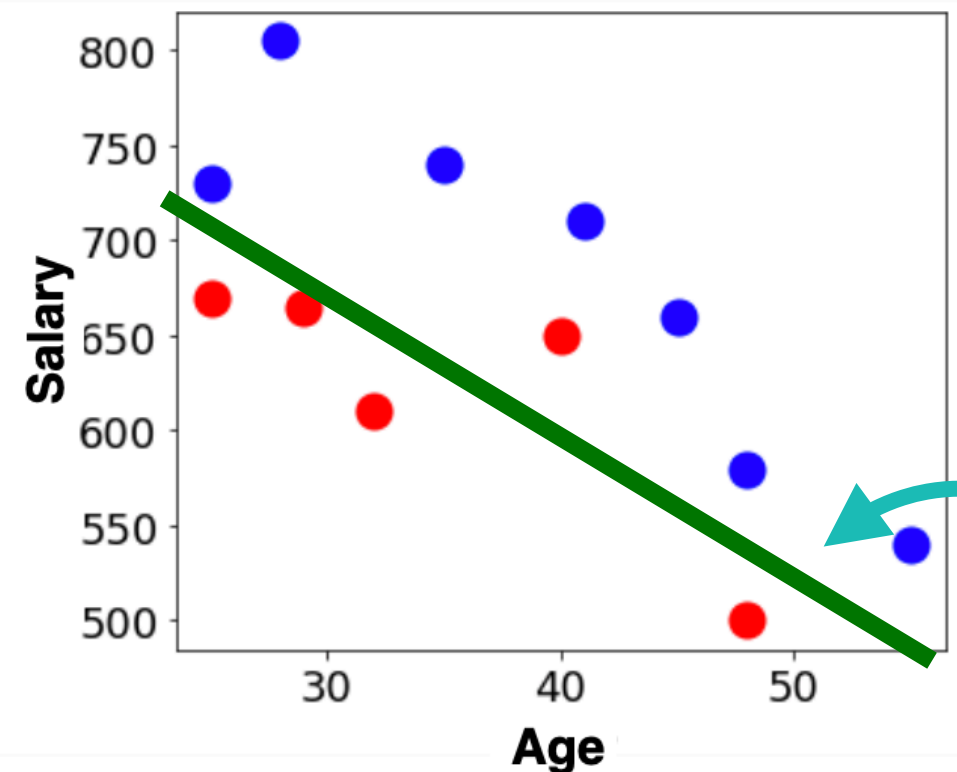
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 - probability that client will Repay Loan
- if $\theta_0 + \theta_1 x_1 + \theta_2 x_2 > 0$
 - predict $y=\text{Yes}$ (Will Repay Loan)
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If $\theta_0 = -3.55$, $\theta_1 = 3.75$, $\theta_2 = 4.82$

What if we need a more complex decision boundary?

Logistic Regression - Decision Boundary

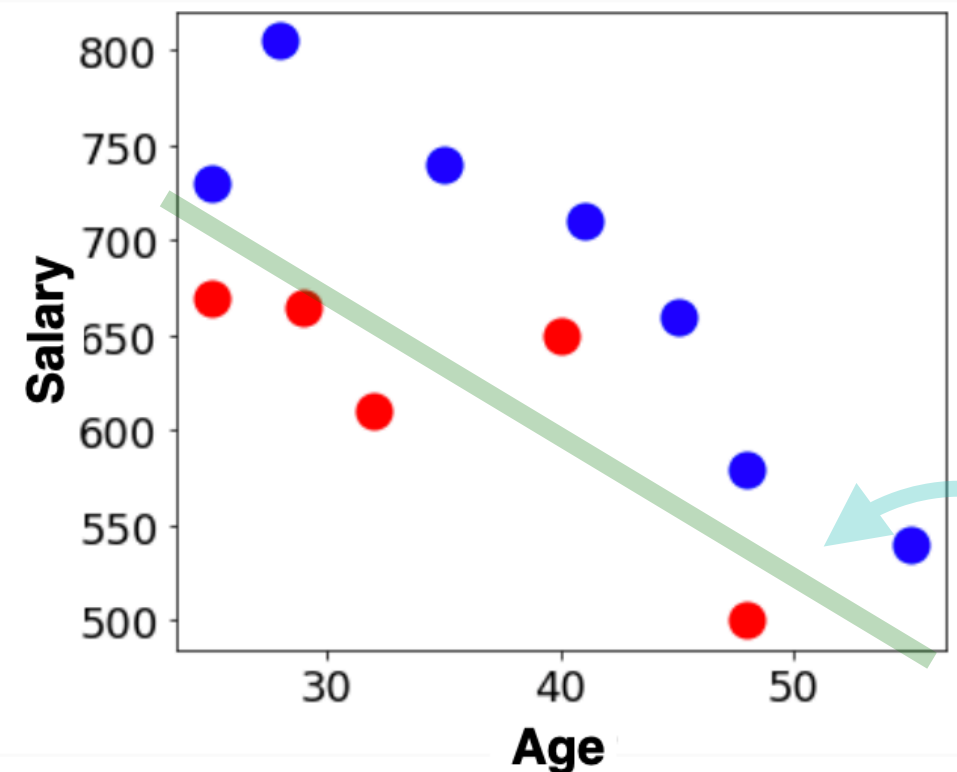
New hypothesis:

$$f(x_1, x_2) = g(\theta_0 + \theta_1(x_1)^2 + \theta_2(x_2)^2)$$

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Idea:

- $f(x_1, x_2)$ is a number between 0-1
- Can be interpreted as a probability
 - probability that client will Repay Loan
- if $\theta_0 + \theta_1(x_1)^2 + \theta_2(x_2)^2 > 0$
 - predict $y=\text{Yes}$ (Will Repay Loan)
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- separates instances with predictions $y=\text{Yes}$ and $y=\text{No}$
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If $\theta_0 = -3.55$, $\theta_1 = 3.75$, $\theta_2 = 4.82$

Logistic Regression - Decision Boundary

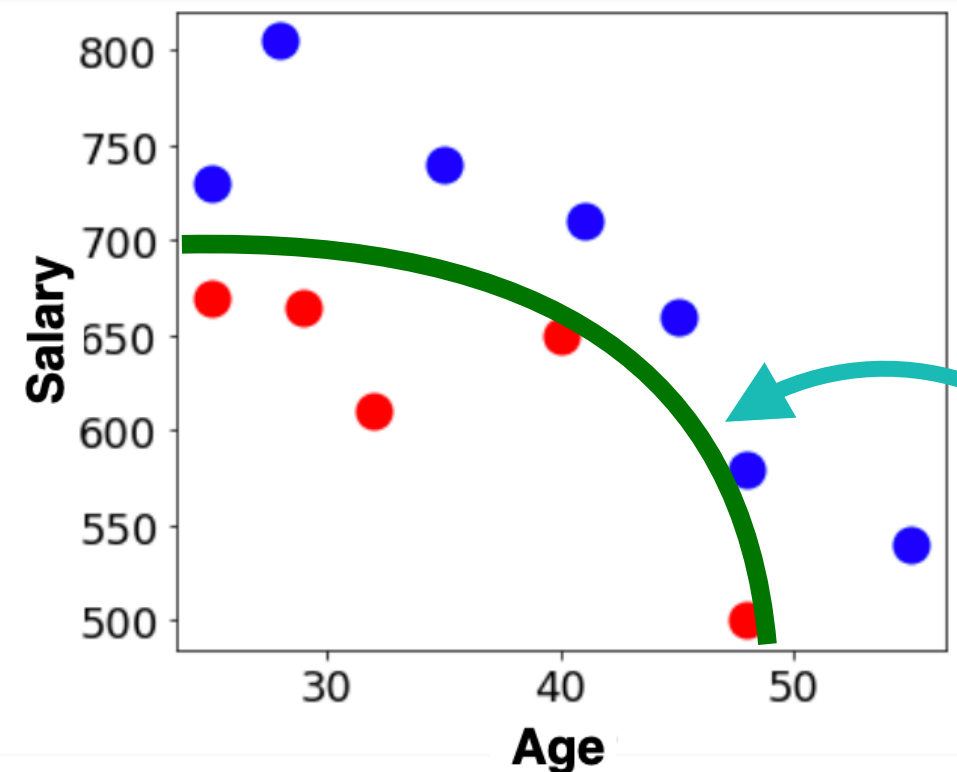
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 - probability that client will Repay Loan
- if $\theta_0 + \theta_1(x_1)^2 + \theta_2(x_2)^2 > 0$
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 - predict $y=\text{No}$ (Will not Repay Loan)



Decision boundary

- separates instances with predictions $y=\text{Yes}$ and $y=\text{No}$
- the surface containing all points (x_1, x_2)
 - for which $\theta_0 + \theta_1(x_1)^2 + \theta_2(x_2)^2 = 0$

If $\theta_0 = -2.68$, $\theta_1 = 4.45$, $\theta_2 = 5.92$

Logistic Regression: Cost Function

Hypothesis:

$$f(x_1, x_2) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

where $g(.)$ is a function that maps arbitrary input values to the $[0-1]$ interval

- $f(x_1, x_2)$ is a number between 0-1
 - probability that the client will Repay Loan

	$x_1 =$ Age	$x_2 =$ Salary	$y =$ Repay Loan
Client1	25	670	No
Client2	29	665	No
Client3	32	610	No
Client4	40	650	No
Client5	48	500	No
Client6	25	730	Yes
Client7	28	805	Yes
Client8	35	740	Yes
Client9	41	710	Yes
Client10	45	660	Yes
Client11	48	580	Yes
Client12	55	540	Yes

How the measure/quantify the performance, J , of one particular logistic regression model?

(i.e., the performance associated with a given set of weights used to make predictions)

Logistic Regression: Cost Function

Hypothesis:

$$f(x_1, x_2) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

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- $f(x_1, x_2)$ is a number between 0-1
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Client8	35	740	Yes
Client9	41	710	Yes
Client10	45	660	Yes
Client11	48	580	Yes
Client12	55	540	Yes

Consider **Client6** ($x_1=25, x_2=730$)

- **Client6** will Repay Loan ($y=Yes$)

Assume $f(25, 730)=1.0$

- what is the error/cost of this prediction?
- predicts that client will Repay Loan with 100% probability!
- **zero** error

Assume $f(25, 730)=0.8$

- what is the error/cost of this prediction?
- predicts that client will Repay Loan with 80% probability
- error is **slightly larger than zero**

Assume $f(25, 730)=0.05$

- what is the error/cost of this prediction?
- predicts that client will Repay Loan with 5% probability
- error is **significantly larger than zero**

In other words, the higher the predicted value $f()$, the smaller the error

Logistic Regression: Cost Function

Hypothesis:

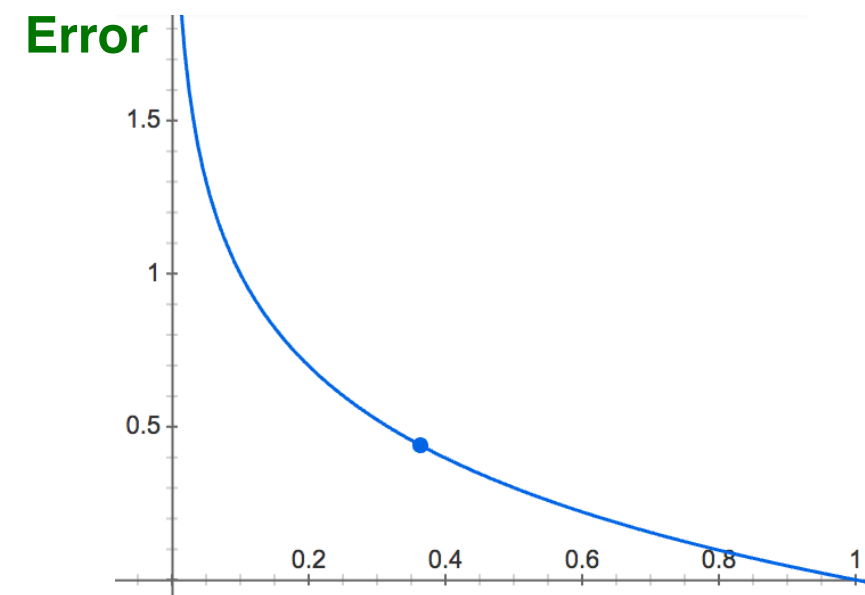
$$f(x_1, x_2) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

where $g(.)$ is a function that maps arbitrary input values to the $[0-1]$ interval

- $f(x_1, x_2)$ is a number between 0-1
 - probability that the client will Repay Loan

	$x_1 =$ Age	$x_2 =$ Salary	$y =$ Repay Loan
Client1	25	670	No
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Client7	28	805	Yes
Client8	35	740	Yes
Client9	41	710	Yes
Client10	45	660	Yes
Client11	48	580	Yes
Client12	55	540	Yes

Consider Client6 ($x_1=25, x_2=730$)
- Client6 will Repay Loan ($y=Yes$)



$f(25, 730)$
(Predicted probability
that they will Repay Loan)

In other words, the higher the predicted value $f()$, the smaller the error

Logistic Regression: Cost Function

Hypothesis:

$$f(x_1, x_2) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

where $g(.)$ is a function that maps arbitrary input values to the $[0-1]$ interval

- $f(x_1, x_2)$ is a number between 0-1
 - probability that the client will Repay Loan



	$x_1 =$ Age	$x_2 =$ Salary	$y =$ Repay Loan
Client1	25	670	No
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Client7	28	805	Yes
Client8	35	740	Yes
Client9	41	710	Yes
Client10	45	660	Yes
Client11	48	580	Yes
Client12	55	540	Yes

Consider **Client4** ($x_1=40, x_2=650$)

- **Client4** will not Repay Loan ($y=\text{No}$)

Assume $f(40, 650)=0.0$

- what is the error/cost of this prediction?
- predicts that client will Repay Loan with 0% probability!
- **zero** error

Assume $f(40, 650)=0.1$

- what is the error/cost of this prediction?
- predicts that client will Repay Loan with 10% probability
- error is **slightly larger than zero**

Assume $f(40, 650)=0.95$

- what is the error/cost of this prediction?
- predicts that client will Repay Loan with 95% probability
- error is **significantly larger than zero**

In other words, the higher the predicted value $f()$, the higher the error

Logistic Regression: Cost Function

Hypothesis:

$$f(x_1, x_2) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

where $g(\cdot)$ is a function that maps arbitrary input values to the $[0-1]$ interval

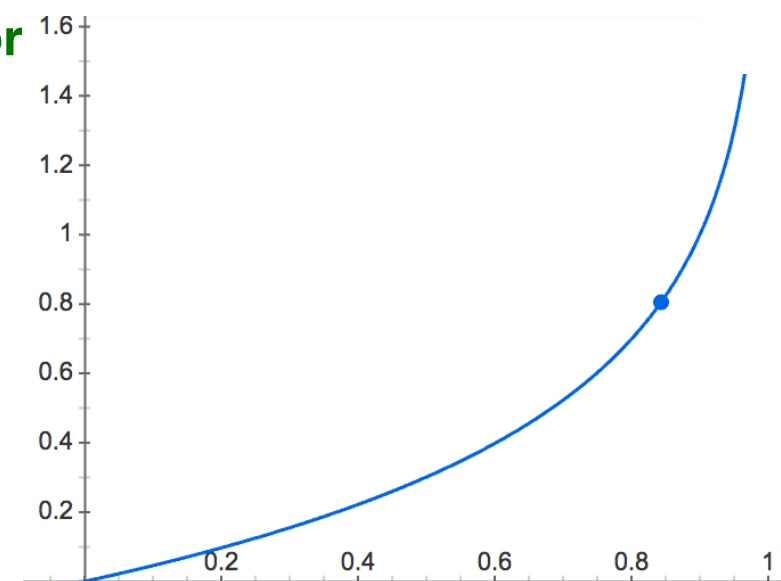
- $f(x_1, x_2)$ is a number between 0-1
 - probability that the client will Repay Loan

	$x_1 =$ Age	$x_2 =$ Salary	$y =$ Repay Loan
Client1	25	670	No
Client2	29	665	No
Client3	32	610	No
Client4	40	650	No
Client5	48	500	No
Client6	25	730	Yes
Client7	28	805	Yes
Client8	35	740	Yes
Client9	41	710	Yes
Client10	45	660	Yes
Client11	48	580	Yes
Client12	55	540	Yes

Consider Client4 ($x_1=40, x_2=650$)

- Client4 will not Repay Loan ($y=\text{No}$)

Error



$f(40, 650)$

(Predicted probability that they will Repay Loan)

In other words, the higher the predicted value $f()$, the higher the error

Logistic Regression: Cost Function

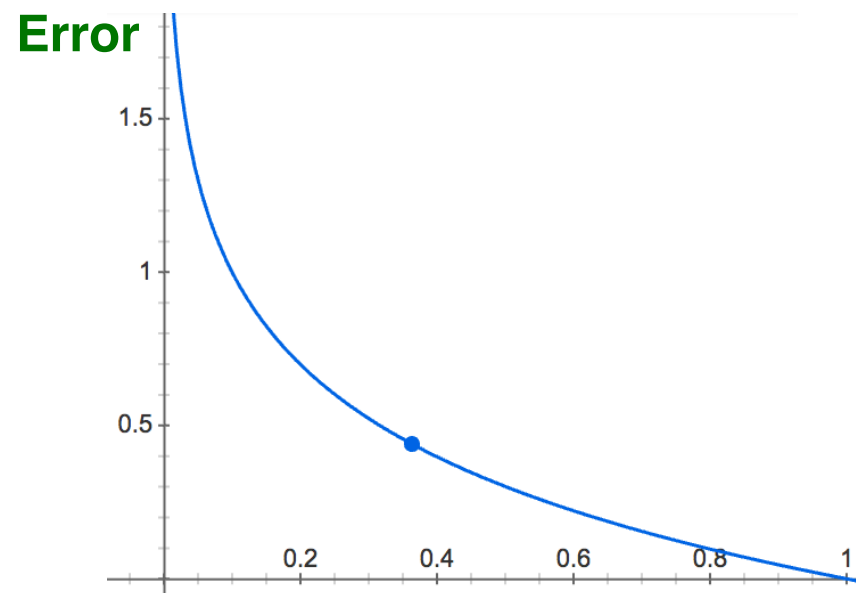
Hypothesis:

$$f(x_1, x_2) = g(\theta_0 + \theta_1 x_1 + \theta_2 x_2)$$

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- $f(x_1, x_2)$ is a number between 0-1
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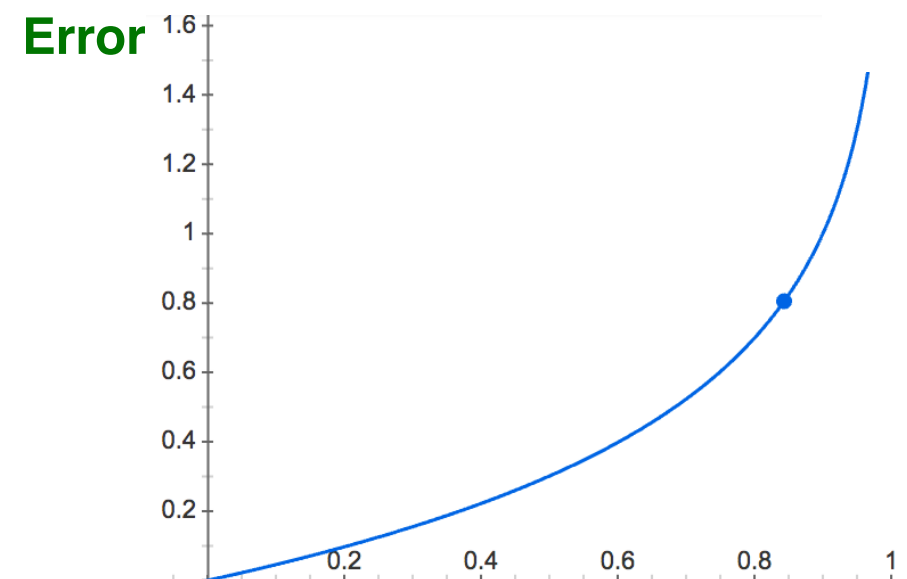
For training instances for which the class is $y=\text{Yes}$ we quantify the prediction error as follows:



$f(25, 730)$
(Predicted probability
that they will Repay Loan)

$$-\log(f_{\theta}(x_1, x_2))$$

For training instances for which the class is $y=\text{No}$ we quantify the prediction error as follows:

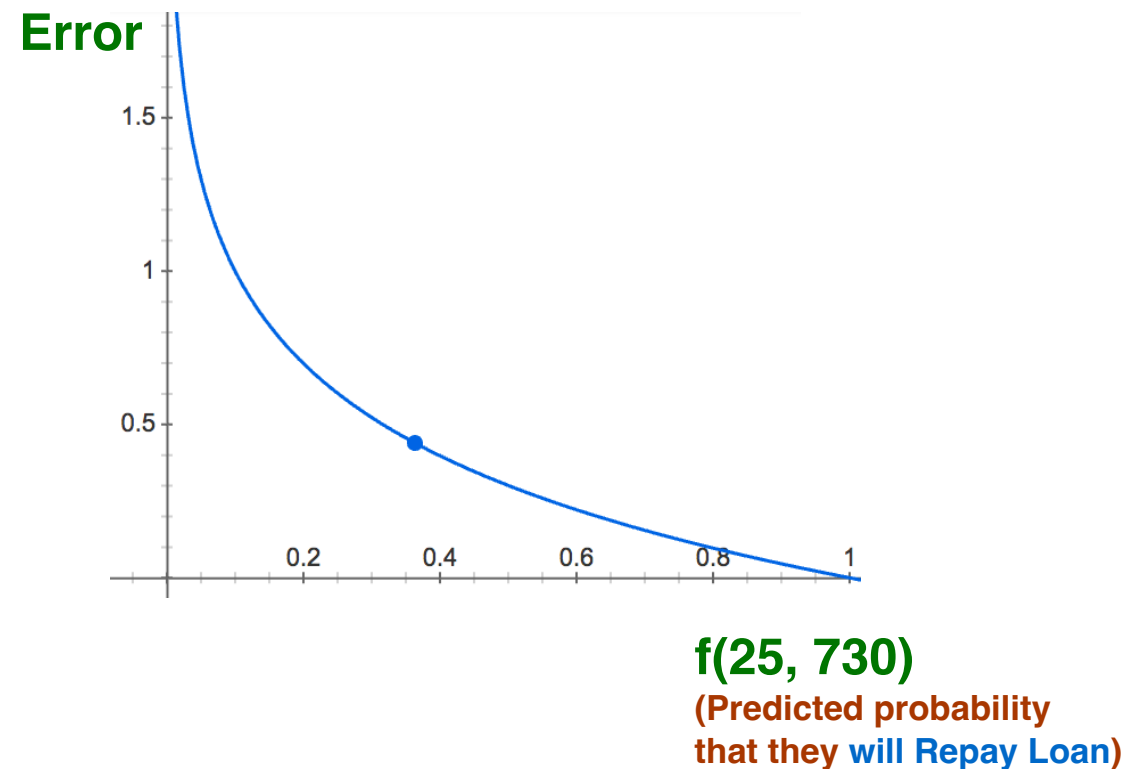


$f(40, 650)$
(Predicted probability
that they will Repay Loan)

$$-\log(1 - f_{\theta}(x_1, x_2))$$

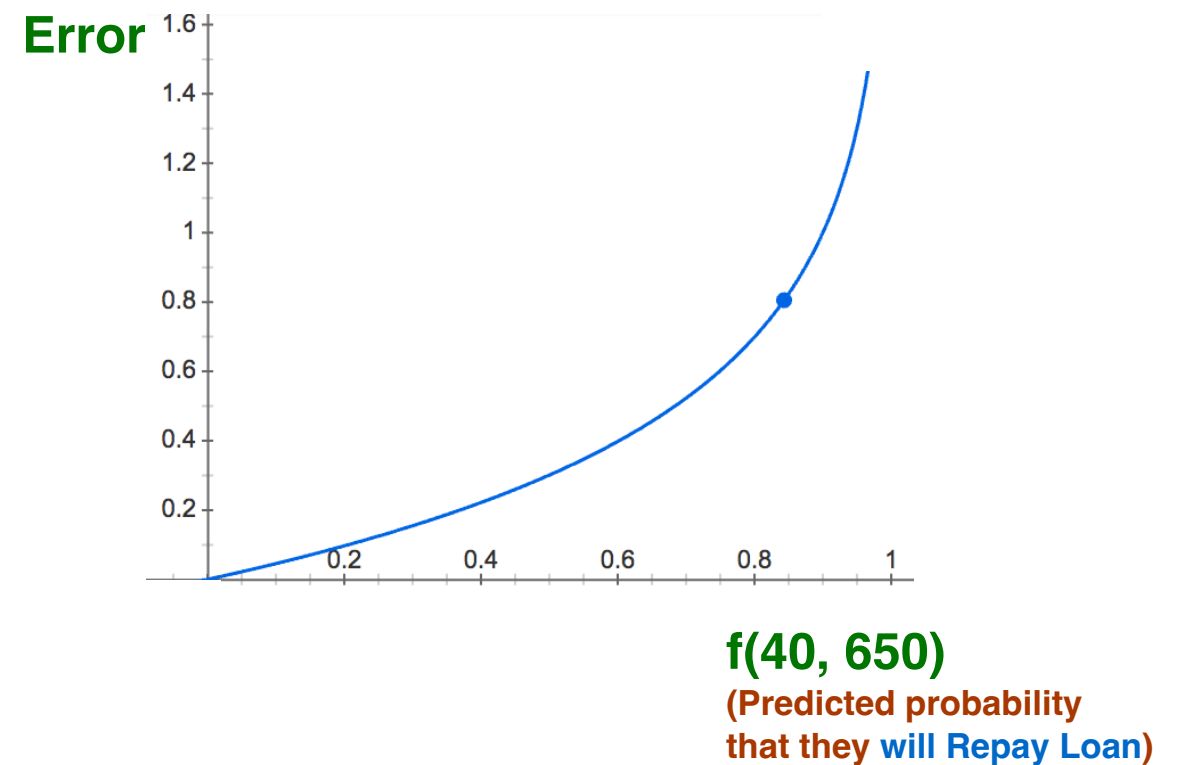
Logistic Regression: Cost Function

For training instances for which the class is **y=Yes** we quantify the prediction error as follows:



$$-\log(f_{\theta}(x_1, x_2))$$

For training instances for which the class is **y=No** we quantify the prediction error as follows:



$$-\log(1 - f_{\theta}(x_1, x_2))$$

$$\text{Error} \left(f_{\theta}(x_1, x_2), y \right) = \begin{cases} -\log(f_{\theta}(x_1, x_2)) & \text{if } y = \text{Yes} \\ -\log(1 - f_{\theta}(x_1, x_2)) & \text{if } y = \text{No} \end{cases}$$

Logistic Regression: Cost Function

$$\begin{aligned}\text{Error} \left(f_{\theta}(x_1, x_2), y \right) &= \begin{cases} -\log(f_{\theta}(x_1, x_2)) & \text{if } y = \text{Yes } (\mathbf{y=1}) \\ -\log(1 - f_{\theta}(x_1, x_2)) & \text{if } y = \text{No } (\mathbf{y=0}) \end{cases} \\ &= y \left(-\log(f_{\theta}(x_1, x_2)) \right) - (1 - y) \left(\log(1 - f_{\theta}(x_1, x_2)) \right)\end{aligned}$$

Logistic Regression: Cost Function

$$\text{Error} \left(f_{\theta}(x_1, x_2), y \right) = \begin{cases} -\log(f_{\theta}(x_1, x_2)) & \text{if } y = \text{Yes } (\mathbf{y=1}) \\ -\log(1 - f_{\theta}(x_1, x_2)) & \text{if } y = \text{No } (\mathbf{y=0}) \end{cases}$$

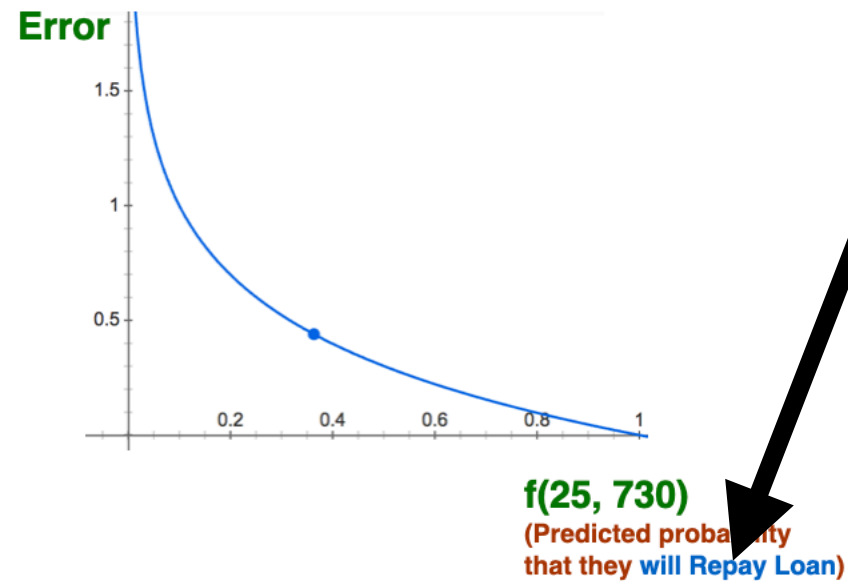
$$= y \left(-\log(f_{\theta}(x_1, x_2)) \right) - \cancel{(1 - y) \left(\log(1 - f_{\theta}(x_1, x_2)) \right)}$$

if y=Yes (1)

Logistic Regression: Cost Function

$$\text{Error} \left(f_{\theta}(x_1, x_2), y \right) = \begin{cases} -\log(f_{\theta}(x_1, x_2)) & \text{if } y = \text{Yes } (y=1) \\ -\log(1 - f_{\theta}(x_1, x_2)) & \text{if } y = \text{No } (y=0) \end{cases}$$
$$= \left(-\log(f_{\theta}(x_1, x_2)) \right)$$

For training instances for which the class is **y=Yes**
we quantify the prediction error as follows:



$$-\log(f_{\theta}(x_1, x_2))$$

if **y=Yes (1)**



=

Logistic Regression: Cost Function

$$\text{Error} \left(f_{\theta}(x_1, x_2), y \right) = \begin{cases} -\log(f_{\theta}(x_1, x_2)) & \text{if } y = \text{Yes } (\mathbf{y=1}) \\ -\log(1 - f_{\theta}(x_1, x_2)) & \text{if } y = \text{No } (\mathbf{y=0}) \end{cases}$$

~~$$= y \left(-\log(f_{\theta}(x_1, x_2)) \right) - (1 - y) \left(\log(1 - f_{\theta}(x_1, x_2)) \right)$$~~

if y=No (0)

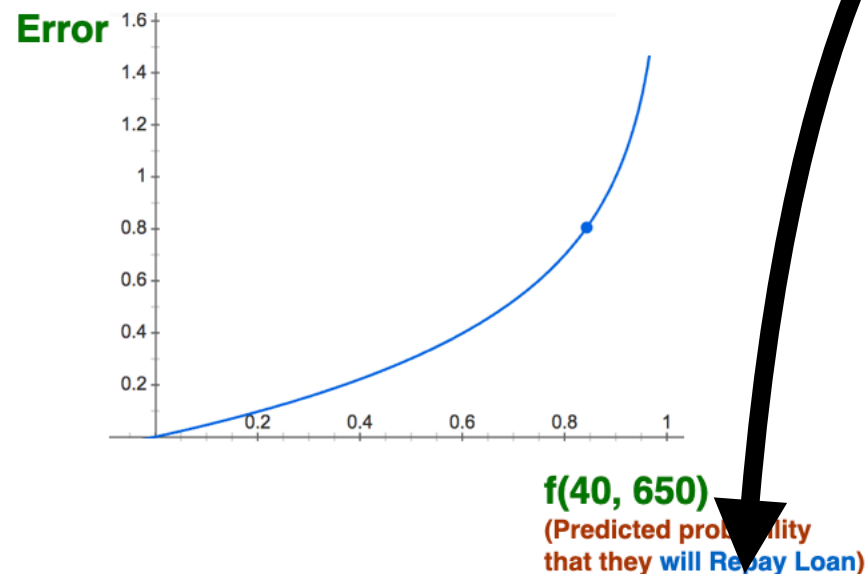
Logistic Regression: Cost Function

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$$= - \left(\log(1 - f_{\theta}(x_1, x_2)) \right)$$

For training instances for which the class is **y=No** we quantify the prediction error as follows:

if y=No (0)



$$-\log(1 - f_{\theta}(x_1, x_2))$$



Logistic Regression: Cost Function

$$\begin{aligned}\text{Error} \left(f_{\theta}(x_1, x_2), y \right) &= \begin{cases} -\log(f_{\theta}(x_1, x_2)) & \text{if } y = \text{Yes } (\mathbf{y=1}) \\ -\log(1 - f_{\theta}(x_1, x_2)) & \text{if } y = \text{No } (\mathbf{y=0}) \end{cases} \\ &= y \left(-\log(f_{\theta}(x_1, x_2)) \right) - (1 - y) \left(\log(1 - f_{\theta}(x_1, x_2)) \right)\end{aligned}$$

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$$\begin{aligned}J(\theta_0, \theta_1, \theta_2) &= \frac{1}{n} \sum_{i=1}^n \text{Error} \left(f_{\theta}(x_1^{(i)}, x_2^{(i)}), y^{(i)} \right) \\ &= \frac{1}{n} \sum_{i=1}^n y^{(i)} \left(-\log(f_{\theta}(x_1^{(i)}, x_2^{(i)})) \right) - (1 - y^{(i)}) \left(\log(1 - f_{\theta}(x_1^{(i)}, x_2^{(i)})) \right)\end{aligned}$$

We wish to find values of θ_0 and θ_1 and θ_2 that minimize the cost function J

Logistic Regression via Gradient Descent

$$J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n -y^{(i)} \left(-\log(f_{\theta}(x_1^{(i)}, x_2^{(i)})) \right) - (1 - y^{(i)}) \left(\log(1 - f_{\theta}(x_1^{(i)}, x_2^{(i)})) \right)$$

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→ To decrease the error J , we should move each θ in the opposite direction of the derivative

Gradient Descent Algorithm

Logistic Regression via Gradient Descent

We wish to find values of θ_0 and θ_1 and θ_2 that minimize the cost function J

$$J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n -y^{(i)} \left(-\log(f_{\theta}(x_1^{(i)}, x_2^{(i)})) \right) - (1 - y^{(i)}) \left(\log(1 - f_{\theta}(x_1^{(i)}, x_2^{(i)})) \right)$$

where $f_{\theta}(x_1, x_2) = g\left(\theta_0 + \theta_1 x_1 + \theta_2 x_2\right) = \frac{1}{1 + \exp\left(-(\theta_0 + \theta_1 x_1 + \theta_2 x_2)\right)}$ is our hypothesis

To decrease the error J , we should move each θ in the opposite direction of the derivative

$$\frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1, \theta_2) \quad \theta_0 : = \theta_0 - \alpha \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1, \theta_2)$$

$$\frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1, \theta_2) \quad \theta_1 : = \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_0, \theta_1, \theta_2)$$

$$\frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2) \quad \theta_2 : = \theta_2 - \alpha \frac{\partial}{\partial \theta_2} J(\theta_0, \theta_1, \theta_2)$$

Logistic Regression via Gradient Descent

We wish to find values of θ_0 and θ_1 and θ_2 that minimize the cost function J

$$J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n -y^{(i)} \left(-\log(f_{\theta}(x_1^{(i)}, x_2^{(i)})) \right) - (1 - y^{(i)}) \left(\log(1 - f_{\theta}(x_1^{(i)}, x_2^{(i)})) \right)$$

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Logistic Regression via Gradient Descent

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$$\begin{aligned}\theta_0 &: = \theta_0 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x_1^{(i)}, x_2^{(i)}) - y^{(i)} \right) \mathbf{1} \\ \theta_1 &: = \theta_1 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x_1^{(i)}, x_2^{(i)}) - y^{(i)} \right) x_1^{(i)} \\ \theta_2 &: = \theta_2 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x_1^{(i)}, x_2^{(i)}) - y^{(i)} \right) x_2^{(i)}\end{aligned}$$

Logistic Regression via Gradient Descent

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Logistic Regression via Gradient Descent

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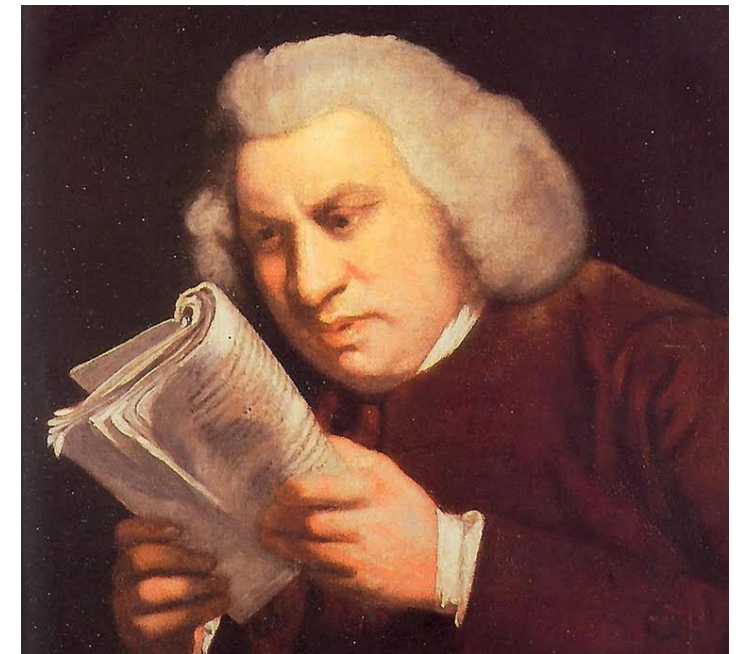
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WHAT

Logistic Regression via Gradient Descent

We wish to find values of θ_0 and θ_1 and θ_2 that minimize the cost function J

$$J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n -y^{(i)} \left(-\log(f_{\theta}(x_1^{(i)}, x_2^{(i)})) \right) - (1 - y^{(i)}) \left(\log(1 - f_{\theta}(x_1^{(i)}, x_2^{(i)})) \right)$$

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} same
update rules
as in
linear regression!

Logistic Regression

Hypothesis: $f_{\theta}(x_1, x_2) = g\left(\theta_0 + \theta_1 x_1 + \theta_2 x_2\right) = \frac{1}{1 + \exp\left(-\left(\theta_0 + \theta_1 x_1 + \theta_2 x_2\right)\right)}$

**Error function
to be minimized:**

$$J(\theta_0, \theta_1, \theta_2) = \frac{1}{n} \sum_{i=1}^n -y^{(i)} \left(\log(f_{\theta}(x_1^{(i)}, x_2^{(i)})) \right) - (1 - y^{(i)}) \left(\log(1 - f_{\theta}(x_1^{(i)}, x_2^{(i)})) \right)$$

**Gradient descent
weight update rule**

$$\begin{aligned}\theta_0 &: = \theta_0 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x_1^{(i)}, x_2^{(i)}) - y^{(i)} \right) \\ \theta_1 &: = \theta_1 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x_1^{(i)}, x_2^{(i)}) - y^{(i)} \right) x_1^{(i)} \\ \theta_2 &: = \theta_2 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x_1^{(i)}, x_2^{(i)}) - y^{(i)} \right) x_2^{(i)}\end{aligned}$$

Logistic Regression

What if we need a more complex decision boundary?

New hypothesis: $f_{\theta}(x_1, x_2) = g\left(\theta_0 + \theta_1(x_1)^2 + \theta_2(x_2)^2\right) = \frac{1}{1 + \exp\left(-(\theta_0 + \theta_1(x_1)^2 + \theta_2(x_2)^2)\right)}$

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$$\begin{aligned}\theta_0 &: = \theta_0 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x_1^{(i)}, x_2^{(i)}) - y^{(i)} \right) 1 \\ \theta_1 &: = \theta_1 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x_1^{(i)}, x_2^{(i)}) - y^{(i)} \right) (x_1^{(i)})^2 \\ \theta_2 &: = \theta_2 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x_1^{(i)}, x_2^{(i)}) - y^{(i)} \right) (x_2^{(i)})^2\end{aligned}$$

Logistic Regression

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$$\theta_2 : = \theta_2 - \alpha \frac{1}{n} \sum_{i=1}^n \left(f_{\theta}(x_1^{(i)}, x_2^{(i)}) - y^{(i)} \right) (x_2^{(i)})^2$$



Machine Learning

CMPSCI 589

Bruno C. da Silva

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Logistic Regression

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College of Information
& Computer Sciences



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